



# Document Object Model

## 1. What is DOM?

DOM stands for **Document Object Model**.

When a browser loads an HTML page, it converts the HTML into a **tree-like structure of objects**.

JavaScript uses this structure to:

- Read content
- Change content
- Add new elements
- Remove elements
- Handle user actions (events)

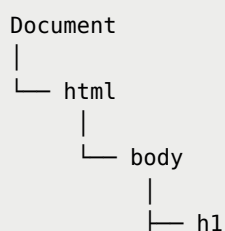
---

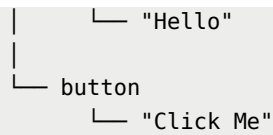
## Real Life Analogy 🏠

Imagine a house.

```
<html>
  <body>
    <h1>Hello</h1>
    <button>Click Me</button>
  </body>
</html>
```

Browser sees it like:





Every tag becomes an object.

This entire structure is called **DOM Tree**.

---

## Why DOM is Required?

Without DOM:

✗ HTML would be static

With DOM:

✓ Change text dynamically

✓ Change styles

✓ Add/Delete elements

✓ Respond to user actions

---

## Example

HTML

```
<h1 id="title">Welcome</h1>
```

JavaScript

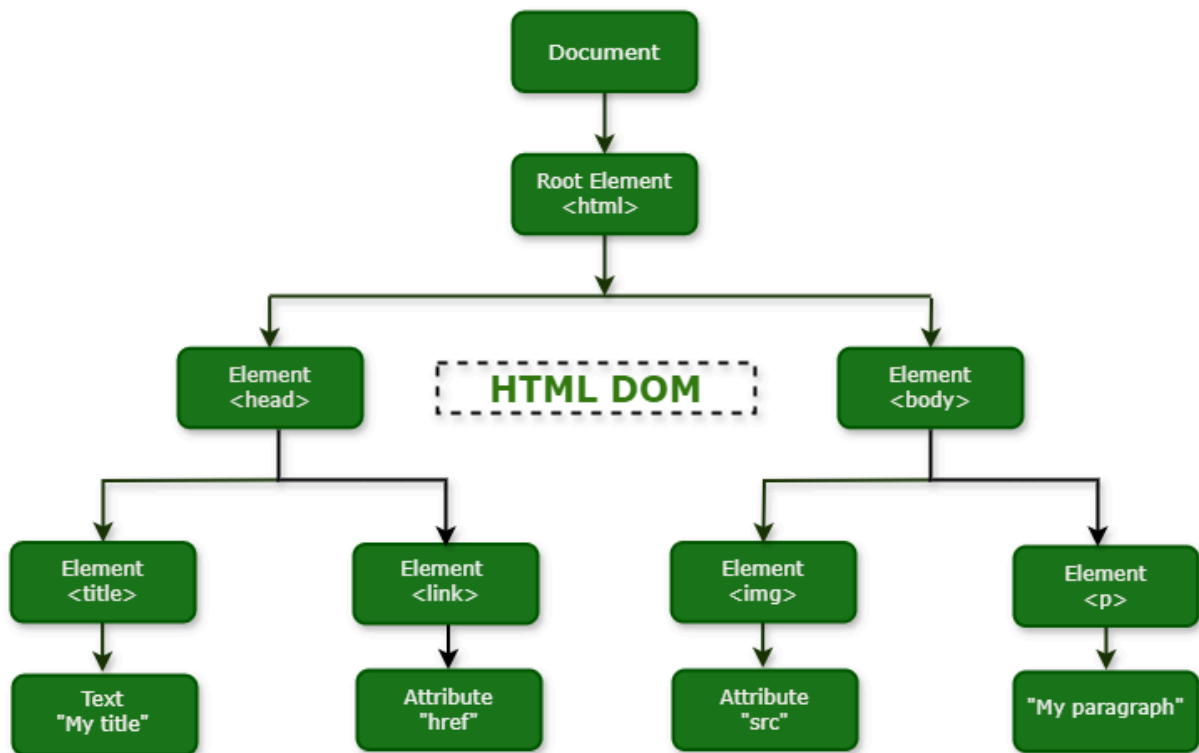
```
document.getElementById("title").innerText = "Hello Students";
```

Output:

Welcome

becomes

## 2. What is Document Object?



The top-most object in DOM is:

document

Think of it as:

```
Document
|
├── HTML
├── HEAD
├── BODY
├── Elements
└── Text
```

Everything starts from document.

Example:

```
console.log(document);
```

Browser prints complete page structure.

---

## Important Document Properties

---

### document.title

Gets or sets page title.

HTML

```
<title>My Website</title>
```

JavaScript

```
console.log(document.title);  
  
document.title = "CodingRank School";
```

### document.URL

```
console.log(document.URL);
```

Output:

```
https://example.com
```

### document.body

Access complete body.

```
console.log(document.body);
```

## document.head

```
console.log(document.head);
```

### 3. Selecting Elements

Before changing anything, JavaScript must find the element.

Think of DOM as a city.

Before visiting a house, you need its address.

Selection methods help find elements.

#### 1. getElementById()

Most commonly used.

HTML

```
<h1 id="heading">Hello</h1>
```

JS

```
var element = document.getElementById("heading");  
  
console.log(element);
```

Output:

```
<h1 id="heading">Hello</h1>
```

#### Change Text

```
document.getElementById("heading").innerText = "Welcome";
```

#### Change Style

```
document.getElementById("heading").style.color = "red";
```

## 2. getElementsByClassName()

Returns multiple elements.

HTML

```
<p class="info">A</p>
<p class="info">B</p>
<p class="info">C</p>
```

JS

```
var elements =
document.getElementsByClassName("info");

console.log(elements);
```

Output:

```
HTMLCollection(3)
```

Access individual item:

```
console.log(elements[0]);
```

## Loop Through

```
var elements =
document.getElementsByClassName("info");

for(var item of elements){
```

```
    console.log(item.innerText);  
}
```

Output:

```
A  
B  
C
```

### 3. getElementsByTagName()

HTML

```
h1  
p  
p  
p
```

JS

```
let para =  
document.getElementsByTagName("p");
```

Returns:

```
HTMLCollection
```

### 4. querySelector()

Modern approach.

Returns first matching element.

HTML

```
<h1 id="title">Hello</h1>
```

JS

```
let element =  
document.querySelector("#title");
```

## Select by Class

```
document.querySelector(".box");
```

## Select by Tag

```
document.querySelector("p");
```

## Nested Selection(Combinator)

```
document.querySelector("div p");
```

First p inside div.

# 5. querySelectorAll()

Returns all matching elements.

HTML

```
<p>One</p>  
<p>Two</p>  
<p>Three</p>
```

JS

```
let items =  
document.querySelectorAll("p");
```

Output:



```
NodeList(3)
```

Loop:

```
items.forEach(item=>{  
  console.log(item.innerText);  
});
```

## Which Approach is Better?

| Method                 | Old/New | Returns        |
|------------------------|---------|----------------|
| getElementById         | Old     | Single Element |
| getElementsByClassName | Old     | HTMLCollection |
| getElementsByTagName   | Old     | HTMLCollection |
| querySelector          | Modern  | First Match    |
| querySelectorAll       | Modern  | NodeList       |

## My Recommendation

Use:

```
querySelector()  
querySelectorAll()
```

Why?

One syntax for everything.

```
#id  
.class  
tag  
div p
```

All possible.

# Text Manipulation

---

## innerText

Gets visible text.

HTML

```
<h1 id="title">Hello</h1>
```

JS

```
console.log(  
  document.getElementById("title").innerText  
);
```

Output:

```
Hello
```

---

## innerHTML

Returns HTML.

HTML

```
<div id="box">  
  <h2>Hello</h2>  
</div>
```

JS

```
console.log(  
  document.getElementById("box").innerHTML  
);
```

Output:

```
<h2>Hello</h2>
```

## textContent

Returns complete text.

```
element.textContent
```

## Attribute Methods

HTML

```

```

## getAttribute()

```
let image =  
document.getElementById("pic");  
  
console.log(  
image.getAttribute("src")  
);
```

Output:

```
cat.jpg
```

## setAttribute()

```
image.setAttribute(  
"src",
```

```
"dog.jpg"  
);
```

## Style Manipulation

HTML

```
<h1 id="title">Hello</h1>
```

JS

```
let title =  
document.getElementById("title");  
  
title.style.color="red";  
title.style.backgroundColor="yellow";  
title.style.fontSize="40px";
```

## classList

Best way to manage CSS classes.

HTML

```
<h1 id="title">Hello</h1>
```

## Add Class

```
title.classList.add("active");
```

## Remove Class

```
title.classList.remove("active");
```

## Toggle

```
title.classList.toggle("dark");
```

Dark mode button often uses toggle.

---

## Contains

```
title.classList.contains("active");
```

Returns:

```
true
```

or

```
false
```

---

## Creating Elements

### createElement()

```
let h1 =  
document.createElement("h1");
```

Creates element in memory.

---

## Add Text

```
h1.innerText =  
"Welcome Students";
```

---

## Append to Page

```
document.body.append(h1);
```

Output:

```
<h1>Welcome Students</h1>
```

## append()

Add at end.

```
parent.append(child);
```

## prepend()

Add at beginning.

```
parent.prepend(child);
```

## remove()

```
element.remove();
```

Deletes element.

# DOM Events

## What is Event?

An action performed by user.

Examples:

- Click
- Double Click

- Hover
  - Typing
  - Submit Form
- 

## Real Life Example

Door Bell = Event

You Press Bell → Event Occurs

Owner Opens Door → Event Handler

Same concept in JavaScript.

---

## Click Event

HTML

```
<button id="btn">
Click Me
</button>
```

JS

```
let btn =
document.getElementById("btn");

btn.addEventListener(
"click",
function(){
    alert("Button Clicked");
});
```

## Why addEventListener?

Best Practice.

Can attach multiple events.

```
btn.addEventListener("click",()=>{
  console.log("One");
});

btn.addEventListener("click",()=>{
  console.log("Two");
});
```

Output:

```
One
Two
```

## Common Events

| Event     | Description   |
|-----------|---------------|
| click     | Button click  |
| dblclick  | Double click  |
| mouseover | Mouse enters  |
| mouseout  | Mouse leaves  |
| keydown   | Key pressed   |
| keyup     | Key released  |
| input     | User types    |
| change    | Value changes |
| submit    | Form submit   |
| load      | Page loaded   |

## Input Event Example

HTML

```
<input id="name">
```



JS

```
let input =  
document.getElementById("name");  
  
input.addEventListener(  
  "input",  
  function(){  
    console.log(input.value);  
  });
```

Every keystroke prints value.

---

## Event Object

```
btn.addEventListener(  
  "click",  
  function(event){  
    console.log(event);  
  });
```

event contains details about action.

---

## Event Target

```
btn.addEventListener(  
  "click",  
  function(event){  
    console.log(event.target);  
  });
```

Output:

Clicked element.

---

## Form Submit Example

## HTML

```
<form id="myForm">

<input type="text">

<button>
Submit
</button>

</form>
```

## JS

```
let form =
document.getElementById("myForm");

form.addEventListener(
"submit",
function(event){

    event.preventDefault();

    console.log("Form Submitted");
});
```

---

# DOM Traversal

Moving around the DOM Tree.

---

## Parent

```
element.parentElement
```

---

## Children

```
element.children
```

## First Child

```
element.firstChild
```

## Last Child

```
element.lastChild
```

## Next Sibling

```
element.nextSibling
```

## Previous Sibling

```
element.previousSibling
```

# Interview Question

## Difference Between innerText and innerHTML

| innerText            | innerHTML             |
|----------------------|-----------------------|
| Works with text      | Works with HTML       |
| Safe                 | Can inject HTML       |
| Returns visible text | Returns complete HTML |

Example:

```
element.innerText =
```

```
"<h1>Hello</h1>";
```

Output:

```
<h1>Hello</h1>
```

```
element.innerHTML =  
"<h1>Hello</h1>";
```

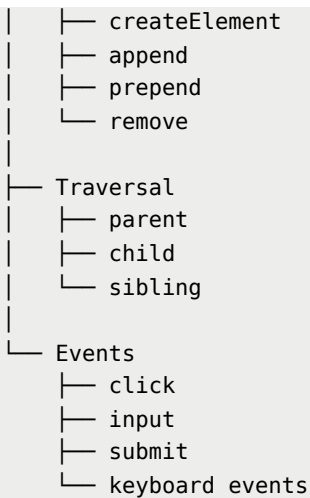
Output:

# Hello

(actual h1 created)

## DOM Learning Roadmap

```
DOM  
├── document  
├── Selection  
│   ├── getElementById  
│   ├── getElementsByClassName  
│   ├── getElementsByTagName  
│   ├── querySelector  
│   └── querySelectorAll  
├── Content  
│   ├── innerText  
│   ├── innerHTML  
│   └── textContent  
├── Attributes  
│   ├── getAttribute  
│   └── setAttribute  
├── Style  
│   └── style property  
├── Classes  
│   └── classList  
└── Create/Delete
```



## Rule for Students

Think of DOM like a **Google Map of your webpage**.

- `document` → Entire city
- `querySelector()` → Search a location
- `innerText` → Change shop name
- `style` → Paint the building
- `createElement()` → Construct new building
- `remove()` → Demolish building
- `addEventListener()` → Add security guard who reacts to actions

### ▼ Student Task

---

## Task 1 - Counter App

### Problem Statement

Instagram likes increase when users click the Like button.

Create a counter application.

When the user clicks:

Increment

count should increase.

When user clicks:

Decrement

count should decrease.

---

## Expected Output

```
0
[ Increment ]
[ Decrement ]

Click Increment

1

Click Increment

2

Click Decrement

1
```

---

## Concepts

```
getElementById()
addEventListener()
innerText
```

---

## HTML + CSS

```
<!DOCTYPE html>
<html>
```

```
<head>
<style>

body{
    height:100vh;
    display:flex;
    justify-content:center;
    align-items:center;
    background:#f4f4f4;
}

.container{
    text-align:center;
}

h1{
    font-size:70px;
    color:#2196f3;
}

button{
    padding:12px 25px;
    margin:10px;
    border:none;
    color:white;
    cursor:pointer;
}

.increment{
    background:green;
}

.decrement{
    background:red;
}

</style>
</head>
```

```
<body>

<div class="container">

  <h1 id="count">0</h1>

  <button
    class="increment"
    id="incrementBtn">

    Increment

  </button>

  <button
    class="decrement"
    id="decrementBtn">

    Decrement

  </button>

</div>

</body>
</html>
```

---

## Task 2 - Live Character Counter

### Problem Statement

WhatsApp shows how many characters you type.

Create the same feature.

---

### Expected Output



Hello

Characters : 5

## Concepts

input event  
value  
innerText

## HTML + CSS

```
<!DOCTYPE html>
<html>
<head>

<style>

.container{
    width:500px;
    margin:50px auto;
}

textarea{
    width:100%;
    height:150px;
    padding:10px;
}

h3{
    margin-top:10px;
}

</style>

</head>
```

```
<body>

<div class="container">

    <textarea
        id="message"
        placeholder="Type Here">
    </textarea>

    <h3>

        Characters :
        <span id="count">0</span>

    </h3>

</div>

</body>
</html>
```

## Task 3 - Show Hide Password

### Problem Statement

Every modern application provides:

 Show Password

feature.

Create it.

### Expected Output

Password

\*\*\*\*\*

Show

↓

Password Visible

---

## Concepts

```
getAttribute()  
setAttribute()  
click event
```

---

## HTML + CSS

```
<!DOCTYPE html>  
<html>  
<head>  
  
<style>  
  
  .container{  
    width:350px;  
    margin:100px auto;  
  }  
  
  input{  
    width:100%;  
    padding:12px;  
  }  
  
  button{  
    width:100%;  
    margin-top:10px;
```

```
padding:12px;
background:#2196f3;
color:white;
border:none;
}

</style>

</head>

<body>

<div class="container">

    <input
        type="password"
        id="password"
        placeholder="Enter Password">

    <button id="toggleBtn">

        Show Password

    </button>

</div>

</body>
</html>
```

## Task 4 - Add Student To List

### Problem Statement

Create a student registration system.

When the user enters:

Rahul

and clicks:

Add Student

student should appear below.

---

## Expected Output

Rahul  
Aman  
Priya

---

## Concepts

```
createElement()  
append()  
value
```

---

## HTML + CSS

```
<!DOCTYPE html>  
<html>  
<head>  
  
<style>  
  
  .container{  
    width:500px;  
    margin:50px auto;  
  }  
  
  input{
```

```

        width:100%;
        padding:12px;
    }

    button{
        width:100%;
        margin-top:10px;
        padding:12px;
    }

    ul{
        margin-top:20px;
    }

    li{
        padding:10px;
        background:#f4f4f4;
        margin-bottom:10px;
    }

</style>

</head>

<body>

<div class="container">

    <input
        id="studentName"
        placeholder="Enter Student Name">

    <button id="addBtn">

        Add Student

    </button>

```

```
<ul id="studentList">

</ul>

</div>

</body>
</html>
```

## Task 5 - To Do App

### Problem Statement

Create a task management application.

Users can:

Add Task  
Delete Task

### Expected Output

Learn JavaScript  
Delete

### Concepts

createElement()  
append()  
remove()  
event.target

## HTML + CSS

```

<!DOCTYPE html>
<html>
<head>

<style>

.container{
    width:600px;
    margin:50px auto;
}

input{
    width:100%;
    padding:12px;
}

button{
    padding:12px;
}

.task{
    display:flex;
    justify-content:space-between;
    background:#f4f4f4;
    padding:15px;
    margin-top:10px;
}

</style>

</head>

<body>

<div class="container">

    <input
        id="task">

```



```
<button
  id="addBtn">

  Add Task

</button>

<div id="taskContainer">

</div>

</div>

</body>
</html>
```

## Task 6 - Employee Registration System

### Problem Statement

Create an Employee Registration Form.

User enters:

Name  
Email  
Country

Click:

Register

Employee details should appear inside a table.

### Concepts

```
Forms  
  
value  
  
createElement()  
  
append()  
  
table row creation
```

## Challenge For Students

```
Easy  
-----  
Counter  
  
Character Counter  
  
Show Password  
  
Medium  
-----  
Student List  
  
To Do App  
  
Hard  
-----  
Employee Registration
```

This sequence gradually teaches students almost every core DOM concept before moving to larger projects like Calculator and Flipkart Clone.

## Good Examples:

### Student Attendance Counter

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
<meta charset="UTF-8">  
<meta name="viewport" content="width=device-width, initial-
```

```

scale=1.0">

<title>Attendance Counter</title>

<style>

    *{
        margin:0;
        padding:0;
        box-sizing:border-box;
    }

    body{
        height:100vh;
        display:flex;
        justify-content:center;
        align-items:center;
        background:#f5f5f5;
        font-family:Arial, sans-serif;
    }

    .container{
        text-align:center;
        background:white;
        padding:30px;
        border-radius:10px;
        box-shadow:0 0 10px rgba(0,0,0,0.2);
    }

    h1{
        font-size:60px;
        color:#2196f3;
        margin-bottom:20px;
    }

    button{
        padding:12px 20px;
        border:none;

```

```

        background:#2196f3;
        color:white;
        border-radius:5px;
        cursor:pointer;
        font-size:16px;
    }

    button:hover{
        background:#1976d2;
    }

</style>

</head>
<body>

<div class="container">

    <h2>Student Attendance Counter</h2>

    <h1 id="count">0</h1>

    <button id="presentBtn">
        Present Student
    </button>

</div>

</body>
</html>

```

## Dark Mode Toggle

```

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">

```

```
<meta name="viewport" content="width=device-width, initial-  
scale=1.0">  
  
<title>Dark Mode Example</title>  
  
<style>  
  
*{  
    margin:0;  
    padding:0;  
    box-sizing:border-box;  
}  
  
body{  
    font-family:Arial, sans-serif;  
    height:100vh;  
    display:flex;  
    justify-content:center;  
    align-items:center;  
    transition:0.3s;  
}  
  
.container{  
    text-align:center;  
}  
  
h1{  
    margin-bottom:20px;  
}  
  
button{  
    padding:12px 20px;  
    border:none;  
    background:#2196f3;  
    color:white;  
    cursor:pointer;  
    border-radius:5px;  
}
```

```

/* Dark Theme */

.dark{
    background:black;
    color:white;
}

</style>

</head>
<body>

<div class="container">

    <h1>Welcome to CodingRank School</h1>

    <button id="themeBtn">
        Toggle Theme
    </button>

</div>

</body>
</html>

<script>

    //Write your code here

</script>

```

## Live Character Counter

```

<!DOCTYPE html>
<html lang="en">
<head>

```

```

<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-
scale=1.0">

<title>Character Counter</title>

<style>

*{
    margin:0;
    padding:0;
    box-sizing:border-box;
}

body{
    height:100vh;
    display:flex;
    justify-content:center;
    align-items:center;
    background:#f4f4f4;
    font-family:Arial, sans-serif;
}

.container{
    width:400px;
    background:white;
    padding:20px;
    border-radius:10px;
    box-shadow:0 0 10px rgba(0,0,0,0.2);
}

h2{
    margin-bottom:15px;
}

textarea{
    width:100%;
    height:150px;

```

```

padding:10px;
resize:none;
font-size:16px;
}

p{
margin-top:10px;
font-size:18px;
}

span{
color:#2196f3;
font-weight:bold;
}

</style>

</head>
<body>

<div class="container">

    <h2>Enter Your Message</h2>

    <textarea id="message"></textarea>

    <p>
        Characters :
        <span id="count">0</span>
    </p>

</div>

</body>
</html>

```

## Add Student to List



```

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-
scale=1.0">

<title>Student Management</title>

<style>

*{
    margin:0;
    padding:0;
    box-sizing:border-box;
}

body{
    font-family:Arial, sans-serif;
    background:#f4f4f4;
    height:100vh;
    display:flex;
    justify-content:center;
    align-items:center;
}

.container{
    width:500px;
    background:white;
    padding:25px;
    border-radius:10px;
    box-shadow:0 0 10px rgba(0,0,0,0.2);
}

h2{
    margin-bottom:20px;
}

```

```

.input-group{
  display:flex;
  gap:10px;
}

input{
  flex:1;
  padding:10px;
  font-size:16px;
}

button{
  padding:10px 20px;
  border:none;
  background:#2196f3;
  color:white;
  cursor:pointer;
  border-radius:5px;
}

ul{
  margin-top:20px;
}

li{
  padding:10px;
  margin-bottom:10px;
  background:#e3f2fd;
  border-radius:5px;
}

</style>

</head>
<body>

<div class="container">

```

```

<h2>Student Management System</h2>

<div class="input-group">

    <input
        type="text"
        id="studentName"
        placeholder="Enter Student Name"
    >

    <button id="addBtn">
        Add Student
    </button>

</div>

<ul id="studentList">

</ul>

</div>

</body>
</html>

```

## Password Eye Example

```

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-
scale=1.0">

<title>Show Hide Password</title>

<!-- Font Awesome -->

```

```
<link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/
6.5.1/css/all.min.css">
```

```
<style>
```

```
*{
    margin:0;
    padding:0;
    box-sizing:border-box;
}

body{
    height:100vh;
    display:flex;
    justify-content:center;
    align-items:center;
    background:#f4f4f4;
    font-family:Arial, sans-serif;
}

.container{
    width:350px;
    background:white;
    padding:25px;
    border-radius:10px;
    box-shadow:0 0 10px rgba(0,0,0,0.2);
}

h2{
    margin-bottom:20px;
    text-align:center;
}

.password-box{
    position:relative;
}
```



```
</div>

<script>

    // fa-eye-slash

</script>

</body>
</html>
```

## Cart Example:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-
scale=1.0">

<title>CodingRank Store</title>

<style>

*{
    margin:0;
    padding:0;
    box-sizing:border-box;
    font-family:Arial, sans-serif;
}

body{
    background:#f4f6f9;
    min-height:100vh;
    padding:40px;
}
```

```
/* Header */

.header{
  display:flex;
  justify-content:space-between;
  align-items:center;
  margin-bottom:40px;
}

.logo{
  font-size:32px;
  font-weight:bold;
  color:#2196f3;
}

.cart{
  background:#2196f3;
  color:white;
  padding:10px 20px;
  border-radius:30px;
  display:flex;
  align-items:center;
  gap:10px;
  font-weight:bold;
  font-size:18px;
}

.cart span{
  width:30px;
  height:30px;
  border-radius:50%;
  background:white;
  color:#2196f3;
  display:flex;
  justify-content:center;
  align-items:center;
}
```

```

/* Products */

.products{
  display:grid;
  grid-template-columns:repeat(auto-fit,minmax(280px,1fr));
  gap:25px;
}

.product-card{
  background:white;
  border-radius:15px;
  overflow:hidden;
  box-shadow:0 5px 15px rgba(0,0,0,0.1);
  transition:0.3s;
}

.product-card:hover{
  transform:translateY(-5px);
}

.product-card img{
  width:100%;
  height:220px;
  object-fit:cover;
}

.product-info{
  padding:20px;
}

.product-info h3{
  margin-bottom:10px;
  color:#333;
}

.product-info p{
  color:#666;
}

```



```

        margin-bottom:15px;
        line-height:1.5;
    }

    .price{
        font-size:24px;
        font-weight:bold;
        color:#2196f3;
        margin-bottom:15px;
    }

    .add-btn{
        width:100%;
        padding:12px;
        border:none;
        border-radius:8px;
        background:#2196f3;
        color:white;
        font-size:16px;
        cursor:pointer;
    }

    .add-btn:hover{
        background:#1976d2;
    }

</style>
</head>
<body>

<!-- Header -->

<div class="header">

    <div class="logo">
         CodingRank Store
    </div>

```

```

    <div class="cart">
      Cart
      <span id="cartCount">0</span>
    </div>

</div>

<!-- Products -->

<div class="products">

  <!-- Product 1 -->

  <div class="product-card">

    <div class="product-info">

      <h3>iPhone 16 Pro</h3>

      <p>
        Latest Apple smartphone with
        powerful performance and camera.
      </p>

      <div class="price">
        ₹89,999
      </div>

      <button class="add-btn">
        Add To Cart
      </button>

    </div>

  </div>

</div>

```

```
<!-- Product 2 -->

<div class="product-card">

    <div class="product-info">

        <h3>Samsung S25 Ultra</h3>

        <p>
            Premium Android flagship with
            AI-powered features.
        </p>

        <div class="price">
            ₹79,999
        </div>

        <button class="add-btn">
            Add To Cart
        </button>

    </div>

</div>

<!-- Product 3 -->

<div class="product-card">

    <div class="product-info">

        <h3>OnePlus 14</h3>

        <p>
```

```

        Fast performance smartphone
        with flagship specifications.
    </p>

    <div class="price">
        ₹64,999
    </div>

    <button class="add-btn">
        Add To Cart
    </button>

</div>

</div>

<!-- Product 4 -->

<div class="product-card">

    <div class="product-info">

        <h3>MacBook Air M4</h3>

        <p>
            Lightweight laptop with
            excellent battery backup.
        </p>

        <div class="price">
            ₹1,09,999
        </div>

        <button class="add-btn">
            Add To Cart
        </button>
    </div>
</div>

```

```

        </div>

    </div>

</div>

</body>
</html>

```

## Calculator Example

```

<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta http-equiv="X-UA-Compatible" content="IE=edge" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Document</title>
    <style>
      h1 {
        text-align: center;
        font-size: 60px;
        color: teal;
      }

      #calculator {
        height: 500px;
        width: 400px;
        box-shadow: rgba(100, 100, 111, 0.2) 0px 7px 29px 0
px;

        margin: auto;
        padding: 40px;
        border-radius: 30px;
      }

```

```

#display {
  height: 60px;
  width: 100%;
  box-shadow: rgba(0, 0, 0, 0.35) 0px 5px 15px;
  margin: auto;
  border-radius: 10px;
}

#keyboard {
  display: grid;
  grid-template-columns: repeat(4, 1fr);
  grid-template-rows: repeat(4, 80px);
  gap: 10px;
  margin-top: 20px;
}

#keyboard > div {
  box-shadow: rgba(50, 50, 93, 0.25) 0px 6px 12px -2px,
    rgba(0, 0, 0, 0.3) 0px 3px 7px -3px;
  border-radius: 10px;
  font-size: 25px;
  font-weight: bold;
  display: flex;
  justify-content: center;
  align-items: center;
}

#keyboard > div:hover {
  background-color: teal;
}

.black {
  background-color: black;
  color: white;
}

.red {

```

```

        background-color: red;
        color: white;
    }
    #display {
        font-size: 30px;
        display: flex;
        align-items: center;
        justify-content: center;
    }
</style>
</head>
<body>
    <h1>Codingrank Calculator</h1>
    <div id="calculator">
        <div id="display"></div>
        <div id="keyboard">
            <div>1</div>
            <div>2</div>
            <div>3</div>
            <div class="black">+</div>
            <div>4</div>
            <div>5</div>
            <div>6</div>
            <div class="black">-</div>
            <div>7</div>
            <div>8</div>
            <div>9</div>
            <div class="black">*</div>
            <div class="red">C</div>
            <div>0</div>
            <div class="black">=</div>
            <div class="black">/</div>
        </div>
    </div>
</body>
</html>

```

# Employee Management System

```
<!DOCTYPE html>

<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-
scale=1.0">

<title>Employee Management System</title>

<style>

*{
    margin:0;
    padding:0;
    box-sizing:border-box;
    font-family:Arial,sans-serif;
}

body{
    background:#f4f7fc;
    padding:30px;
}

.container{
    max-width:1400px;
    margin:auto;
}

.heading{
    text-align:center;
    margin-bottom:30px;
}

.heading h1{
    color:#2563eb;
```



```

        margin-bottom:10px;
    }

    .heading p{
        color:#666;
    }

    /* FORM */

    .form-container{
        background:white;
        padding:25px;
        border-radius:15px;
        box-shadow:0 5px 15px rgba(0,0,0,0.1);
        margin-bottom:30px;
    }

    .form-grid{
        display:grid;
        grid-template-columns:repeat(auto-fit,minmax(250px,1fr));
        gap:20px;
    }

    .input-group{
        display:flex;
        flex-direction:column;
    }

    label{
        margin-bottom:6px;
        font-weight:bold;
        color:#333;
    }

    input,
    select{
        padding:12px;
    }

```

```
    border:1px solid #ddd;
    border-radius:8px;
    font-size:15px;
}

.skills{
    display:flex;
    gap:15px;
    flex-wrap:wrap;
    margin-top:5px;
}

.skills label{
    font-weight:normal;
}

.gender{
    display:flex;
    gap:20px;
    margin-top:5px;
}

.gender label{
    font-weight:normal;
}

.register-btn{
    margin-top:25px;
    width:100%;
    padding:14px;
    border:none;
    background:#2563eb;
    color:white;
    font-size:16px;
    border-radius:8px;
    cursor:pointer;
}
```

```

.register-btn:hover{
    background:#1d4ed8;
}

/* TABLE */

.table-container{
    background:white;
    border-radius:15px;
    overflow:hidden;
    box-shadow:0 5px 15px rgba(0,0,0,0.1);
}

.table-header{
    background:#2563eb;
    color:white;
    padding:18px;
    font-size:20px;
    font-weight:bold;
}

.table-wrapper{
    overflow-x:auto;
}

table{
    width:100%;
    border-collapse:collapse;
}

th{
    background:#eff6ff;
    color:#1e3a8a;
}

th,
td{
    padding:15px;
}

```

```
        text-align:left;
        border-bottom:1px solid #ddd;
    }

    tr:hover{
        background:#f9fafb;
    }

    .action-buttons{
        display:flex;
        gap:10px;
    }

    .edit-btn{
        background:#10b981;
        color:white;
        border:none;
        padding:8px 15px;
        border-radius:6px;
        cursor:pointer;
    }

    .delete-btn{
        background:#ef4444;
        color:white;
        border:none;
        padding:8px 15px;
        border-radius:6px;
        cursor:pointer;
    }

    .edit-btn:hover{
        background:#059669;
    }

    .delete-btn:hover{
        background:#dc2626;
    }
```

```

/* MOBILE CARDS */

#employeeCards{
    display:none;
    padding:20px;
}

.employee-card{
    background:#ffffff;
    border-radius:12px;
    padding:18px;
    margin-bottom:15px;
    box-shadow:0 4px 12px rgba(0,0,0,0.1);
}

.employee-card h3{
    color:#2563eb;
    margin-bottom:10px;
}

.employee-card p{
    margin-bottom:8px;
    color:#444;
}

.card-actions{
    display:flex;
    gap:10px;
    margin-top:15px;
}

/* RESPONSIVE */

@media(max-width:768px){

    body{
        padding:15px;
    }
}

```

```
}

.form-grid{
  grid-template-columns:1fr;
}

.skills{
  flex-direction:column;
  gap:8px;
}

.gender{
  flex-direction:column;
  gap:8px;
}

.table-wrapper{
  display:none;
}

#employeeCards{
  display:block;
}

.card-actions{
  flex-direction:column;
}

.edit-btn,
.delete-btn{
  width:100%;
}

}

</style>

</head>
```

```

<body>

<div class="container">

    <div class="heading">

        <h1>Employee Management System</h1>

        <p>
            Register employees and manage records using JavaScript DOM
        </p>

    </div>

    <!-- FORM -->

    <div class="form-container">

        <form id="employeeForm">

            <div class="form-grid">

                <div class="input-group">
                    <label>Name</label>
                    <input
                        type="text"
                        id="name"
                        placeholder="Enter Employee Name">
                </div>

                <div class="input-group">
                    <label>Email</label>
                    <input
                        type="email"
                        id="email"
                        placeholder="Enter Email">
            </div>
        </form>
    </div>

```

```

        </div>

        <div class="input-group">
            <label>Mobile Number</label>
            <input
                type="number"
                id="mobile"
                placeholder="Enter Mobile Number">
        </div>

        <div class="input-group">
            <label>Country</label>
            <select id="country">
                <option value="">Select Country</option>

                <option>India</option>
                <option>USA</option>
                <option>Canada</option>
                <option>Germany</option>
                <option>Australia</option>
            </select>
        </div>

    </div>

    <br>

    <div class="input-group">

        <label>Skills</label>

        <div class="skills">

            <label>
                <input type="checkbox" value="Java">
                Java
            </label>

```



```

        <label>
            <input type="checkbox" value="Spring Boot">
                Spring Boot
        </label>

        <label>
            <input type="checkbox" value="React">
                React
        </label>

        <label>
            <input type="checkbox" value="Node.js">
                Node.js
        </label>

        <label>
            <input type="checkbox" value="AWS">
                AWS
        </label>

    </div>

</div>

<br>

<div class="input-group">

    <label>Gender</label>

    <div class="gender">

        <label>
            <input
                type="radio"
                name="gender"
                value="Male">

```

```

        Male
    </label>

    <label>
        <input
            type="radio"
            name="gender"
            value="Female">
        Female
    </label>

    <label>
        <input
            type="radio"
            name="gender"
            value="Other">
        Other
    </label>

</div>

</div>

<button
    type="submit"
    class="register-btn">

    Register Employee

</button>

</form>

</div>

<!-- EMPLOYEE RECORDS -->

<div class="table-container">

```

```

<div class="table-header">
    Employee Records
</div>

<!-- Desktop / Tablet Table -->

<div class="table-wrapper">

    <table>

        <thead>

            <tr>

                <th>Name</th>
                <th>Email</th>
                <th>Mobile</th>
                <th>Country</th>
                <th>Gender</th>
                <th>Skills</th>
                <th>Actions</th>

            </tr>

        </thead>

        <tbody id="employeeTableBody">

            <tr>

                <td>Rahul Sharma</td>
                <td>rahul@gmail.com</td>
                <td>9876543210</td>
                <td>India</td>
                <td>Male</td>
                <td>Java, Spring Boot</td>
            </tr>
        </tbody>
    </table>
</div>

```

```

        <td>

            <div class="action-buttons">

                <button class="edit-btn">
                    Edit
                </button>

                <button class="delete-btn">
                    Delete
                </button>

            </div>

        </td>

    </tr>

</tbody>

</table>

</div>

<!-- Mobile Cards -->

<div id="employeeCards">

    <div class="employee-card">

        <h3>Rahul Sharma</h3>

        <p><strong>Email:</strong> rahul@gmail.com</p>

        <p><strong>Mobile:</strong> 9876543210</p>

        <p><strong>Country:</strong> India</p>

```

```
<p><strong>Gender:</strong> Male</p>

<p><strong>Skills:</strong> Java, Spring Boot</p>

<div class="card-actions">

    <button class="edit-btn">
        Edit
    </button>

    <button class="delete-btn">
        Delete
    </button>

</div>

</div>

</div>

</div>

</div>

</body>
</html>
```

## Product Page:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-
scale=1.0">
<title>Flipkart Product Page</title>
```

```

<style>

*{
    margin:0;
    padding:0;
    box-sizing:border-box;
    font-family:Arial, sans-serif;
}

body{
    background:#f1f3f6;
}

/* HEADER */

.header{
    background:#2874f0;
    color:white;
    padding:15px 30px;
    display:flex;
    justify-content:space-between;
    align-items:center;
    flex-wrap:wrap;
    gap:15px;
}

.logo{
    font-size:28px;
    font-weight:bold;
}

.search-box{
    flex:1;
    max-width:500px;
}

.search-box input{

```

```

    width:100%;
    padding:12px;
    border:none;
    border-radius:5px;
    outline:none;
}

.cart{
    background:white;
    color:#2874f0;
    padding:10px 20px;
    border-radius:30px;
    font-weight:bold;
}

/* MAIN */

.main{
    display:flex;
    gap:20px;
    padding:20px;
}

/* SIDEBAR */

.sidebar{
    width:280px;
    background:white;
    padding:20px;
    border-radius:10px;
    box-shadow:0 2px 10px rgba(0,0,0,0.1);
    height:fit-content;
}

.sidebar h2{
    margin-bottom:20px;
    color:#2874f0;
}

```

```

.filter-group{
    margin-bottom:25px;
}

.filter-group h3{
    margin-bottom:10px;
}

.filter-group label{
    display:block;
    margin-bottom:10px;
    cursor:pointer;
}

/* CONTENT */

.content{
    flex:1;
}

.top-bar{
    background:white;
    padding:15px 20px;
    border-radius:10px;
    display:flex;
    justify-content:space-between;
    align-items:center;
    margin-bottom:20px;
    box-shadow:0 2px 10px rgba(0,0,0,0.1);
}

.product-count{
    font-size:18px;
    font-weight:bold;
}

.sort select{

```



```

        padding:10px;
        border:1px solid #ddd;
        border-radius:5px;
    }

    /* PRODUCTS */

    .products{
        display:grid;
        grid-template-columns:
            repeat(auto-fill,minmax(250px,1fr));
        gap:20px;
    }

    .product-card{
        background:white;
        border-radius:10px;
        overflow:hidden;
        box-shadow:0 2px 10px rgba(0,0,0,0.1);
        transition:0.3s;
    }

    .product-card:hover{
        transform:translateY(-5px);
    }

    .product-card img{
        width:100%;
        height:220px;
        object-fit:cover;
    }

    .product-info{
        padding:15px;
    }

    .product-title{
        font-size:18px;
    }

```

```
    font-weight:bold;
    margin-bottom:10px;
}

.category{
    color:#777;
    margin-bottom:10px;
}

.price{
    color:#388e3c;
    font-size:22px;
    font-weight:bold;
    margin-bottom:10px;
}

.rating{
    display:inline-block;
    background:#388e3c;
    color:white;
    padding:4px 8px;
    border-radius:5px;
    margin-bottom:15px;
}

.add-cart{
    width:100%;
    border:none;
    padding:12px;
    background:#ff9f00;
    color:white;
    border-radius:5px;
    cursor:pointer;
    font-size:16px;
}

.add-cart:hover{
    background:#fb8c00;
```

```

}

/* RESPONSIVE */

@media(max-width:900px){

    .main{
        flex-direction:column;
    }

    .sidebar{
        width:100%;
    }

}

@media(max-width:600px){

    .header{
        flex-direction:column;
        align-items:stretch;
    }

    .search-box{
        max-width:100%;
    }

    .top-bar{
        flex-direction:column;
        gap:10px;
        align-items:flex-start;
    }

    .products{
        grid-template-columns:1fr;
    }

}

```

```

</style>
</head>

<body>

<header class="header">

    <div class="logo">
         Flipkart Clone
    </div>

    <div class="search-box">

        <input
            type="text"
            id="search"
            placeholder="Search Products">

    </div>

    <div class="cart">

        Cart :
        <span id="cartCount">0</span>

    </div>

</header>

<div class="main">

    <!-- FILTERS -->

    <aside class="sidebar">

        <h2>Filters</h2>

```

```

<div class="filter-group">

  <h3>Category</h3>

  <label>
    <input
      type="checkbox"
      value="Mobile">
    Mobile
  </label>

  <label>
    <input
      type="checkbox"
      value="Laptop">
    Laptop
  </label>

  <label>
    <input
      type="checkbox"
      value="Watch">
    Watch
  </label>

</div>

<div class="filter-group">

  <h3>Rating</h3>

  <label>
    <input
      type="radio"
      name="rating"
      value="4">
    4★ & Above
  </label>

```

```

        <label>
            <input
                type="radio"
                name="rating"
                value="4.5">
            4.5★ & Above
        </label>

    </div>

</aside>

<!-- PRODUCTS SECTION -->

<section class="content">

    <div class="top-bar">

        <div class="product-count">

            Total Products :
            <span id="count">0</span>

        </div>

        <div class="sort">

            <select id="sort">

                <option value="">
                    Sort By
                </option>

                <option value="asc">
                    Price Low To High
                </option>
            </select>
        </div>
    </div>

```

```

        <option value="desc">
            Price High To Low
        </option>

        <option value="rating">
            Rating High To Low
        </option>

    </select>

</div>

</div>

<!-- JavaScript Will Render Products Here -->

<div
    id="products"
    class="products">

</div>

</section>

</div>

</body>
</html>

```

Javascript:

```

<script>

    // =====
    // PRODUCT DATA
    // =====

    let products = [

```

```
{
  id: 1,
  title: "iPhone 16 Pro",
  category: "Mobile",
  price: 89999,
  rating: 4.8,
  image: "https://picsum.photos/400/300?random=1"
},
{
  id: 2,
  title: "Samsung S25 Ultra",
  category: "Mobile",
  price: 79999,
  rating: 4.7,
  image: "https://picsum.photos/400/300?random=2"
},
{
  id: 3,
  title: "MacBook Air M4",
  category: "Laptop",
  price: 109999,
  rating: 4.9,
  image: "https://picsum.photos/400/300?random=3"
},
{
  id: 4,
  title: "OnePlus 14",
  category: "Mobile",
  price: 64999,
  rating: 4.6,
  image: "https://picsum.photos/400/300?random=4"
},
{
  id: 5,
  title: "Apple Watch",
  category: "Watch",
  price: 45999,
  rating: 4.5,
```



```

        image: "https://picsum.photos/400/300?random=5"
    },
    {
        id: 6,
        title: "Dell XPS 15",
        category: "Laptop",
        price: 129999,
        rating: 4.8,
        image: "https://picsum.photos/400/300?random=6"
    }
];

// =====
// DOM ELEMENTS
// =====

let productsContainer =
document.getElementById("products");

let productCount =
document.getElementById("count");

let searchInput =
document.getElementById("search");

let sortSelect =
document.getElementById("sort");

let categoryCheckboxes =
document.querySelectorAll(
    ".sidebar input[type='checkbox']"
);

let ratingRadios =
document.querySelectorAll(
    "input[name='rating']"
);

```

```

let cartCount =
document.getElementById("cartCount");

// =====
// CART
// =====

let cart = 0;

// =====
// CURRENT FILTERED DATA
// =====

let filteredProducts = [...products];

// =====
// DISPLAY PRODUCTS
// =====

function displayProducts(data){

    productsContainer.innerHTML = "";

    productCount.innerText = data.length;

    data.forEach(function(product){

        let card =
document.createElement("div");

        card.classList.add("product-card");

        card.innerHTML = `

            <div class="product-info">

```

```

        <div class="product-title">
            ${product.title}
        </div>

        <div class="category">
            ${product.category}
        </div>

        <div class="price">
            ₹${product.price}
        </div>

        <div class="rating">
            ${product.rating} ★
        </div>

        <button class="add-cart">
            Add To Cart
        </button>

    </div>

    `;

    productsContainer.append(card);

});

}

// =====
// SEARCH
// =====

searchInput.addEventListener(
    "input",
    function(){

```

```

        applyFilters();

    }

);

// =====
// CATEGORY FILTER
// =====

categoryCheckboxes.forEach(function(box){

    box.addEventListener(
        "change",
        applyFilters
    );

});

// =====
// RATING FILTER
// =====

ratingRadios.forEach(function(radio){

    radio.addEventListener(
        "change",
        applyFilters
    );

});

// =====
// SORT
// =====

sortSelect.addEventListener(
    "change",

```

```

        applyFilters
    );

    // =====
    // APPLY FILTERS
    // =====

    function applyFilters(){

        let result = [...products];

        // SEARCH

        let searchValue =
            searchInput.value.toLowerCase();

        if(searchValue){

            result = result.filter(function(product){

                return product.title
                    .toLowerCase()
                    .includes(searchValue);

            });

        }

        // CATEGORY FILTER

        let selectedCategories = [];

        categoryCheckboxes.forEach(function(box){

            if(box.checked){

                selectedCategories.push(
                    box.value
                )
            }
        })
    }

```

```

        );

    }

});

if(selectedCategories.length > 0){

    result = result.filter(function(product){

        return selectedCategories.includes(
            product.category
        );

    });

}

// RATING FILTER

let selectedRating = null;

ratingRadios.forEach(function(radio){

    if(radio.checked){

        selectedRating =
            Number(radio.value);

    }

});

if(selectedRating){

    result = result.filter(function(product){

        return product.rating >=

```

```
        selectedRating;

    });

}

// SORTING

if(sortSelect.value === "asc"){

    result.sort(function(a,b){

        return a.price - b.price;

    });

}

else if(sortSelect.value === "desc"){

    result.sort(function(a,b){

        return b.price - a.price;

    });

}

else if(sortSelect.value === "rating"){

    result.sort(function(a,b){

        return b.rating - a.rating;

    });

}
```

```

        filteredProducts = result;

        displayProducts(result);
    }

    // =====
    // ADD TO CART
    // =====

    productsContainer.addEventListener(
        "click",
        function(event){

            if(
                event.target.classList.contains(
                    "add-cart"
                )
            ){

                cart++;

                cartCount.innerText = cart;

                let productName =
                event.target
                    .parentElement
                    .querySelector(".product-title")
                    .innerText;

                alert(
                    productName +
                    " added to cart"
                );

            }

        }
    )
}

```



```
);  
  
// =====  
// INITIAL RENDER  
// =====  
  
displayProducts(products);  
  
</script>
```